

Sai Karthik

Business Analyst

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Summary

Results-driven Business Analyst with 4+ years of experience designing scalable cloud-native solutions and building robust data platforms in finance and retail domains. Expert in Python, SQL, Azure, Power BI, and RDBMS with a proven track record of improving operational efficiency by up to 40%, reducing data defects by 25%, and enhancing customer engagement. Skilled in data modeling, ETL pipeline development, and compliance-driven reporting (SOX, HIPAA) to drive actionable business insights.

Education

University at Albany

Albany, NY

Master of Science in Data Science

08/2023 - 05/2025

Skills

Programming & Data Processing: Python (Pandas, NumPy, PySpark), SQL, R, Shell Scripting, Data Cleaning & Transformation, Data Pipeline Development

Data Modeling & Warehousing: Oracle, SAP, MySQL, PostgreSQL, BigQuery, AWS RDS, Data Warehousing (Star & Snowflake Schemas), ETL/ELT Development, Data Lineage & Metadata Management, Schema Design & Optimization

Cloud Platforms & Automation: AWS (S3, Lambda, Glue, Step Functions, Redshift), GCP (BigQuery, Cloud Functions, Dataflow), Azure (Data Factory, Synapse), Alteryx, Airflow, dbt, Workflow Orchestration, Batch & Streaming Data Processing

Big Data & Processing Frameworks: Spark (PySpark), Kafka, Flink, Data Lake Architectures, Distributed Computing, Streaming Ingestion.

Data Quality & Governance: Data Validation & Reconciliation, Data Auditing, Data Cataloging, Governance Frameworks, Regulatory & Compliance Reporting (SOX, HIPAA), Documentation Standards

Analytics & Business Impact: Predictive Analytics, Statistical Analysis, Forecasting, A/B Testing, Business Requirements Gathering, Risk & Audit Analytics, Customer Behavior & Churn Analysis, Revenue Forecasting

Visualization & BI Tools: Tableau, Power BI, Looker, KPI Design & Tracking, Dashboard Development, Data Storytelling.

Collaboration & Tools: Git, GitHub, Jira, Confluence, SharePoint, Microsoft Teams, Slack, Agile/ Scrum Methodologies.

Experience

BNY Mellon

Jersey City, NJ

Senior Business Analyst

07/2024 - Present

- Designed and remodeled logical and physical RDBMS data models for transactional and analytical documented banking systems, supporting six or more applications processing over 12 million daily client, entitlement, and financial transactions.
- Designed and remodeled logical and physical data models for Microsoft SQL Server and Oracle, supporting multiple banking applications processing over 10 million daily transactions across client, entitlement, and financial domains.
- Enhanced enterprise canonical data models to integrate more than three new business capabilities, ensuring schema consistency and interoperability across upstream and downstream RDBMS systems.
- Developed schema extensions to comply with regulatory and HIPAA reporting requirements, reducing manual reconciliation efforts by 30%.
- Optimized database views, stored procedures, and T-SQL scripts, improving query performance by 40% for high-volume OLTP and OLAP workloads.
- Produced and maintained metadata documentation, enabling more than five application teams to efficiently onboard and access platform data.
- Collaborated with analytics and development teams to tune indexing, partitioning, and SSAS cubes, ensuring SLA adherence for reporting and high-volume queries.
- Implemented automated data validation, quality checks, and verification pipelines at key interfaces, reducing production data defects by 25%.
- Conducted cross-functional walkthroughs and training sessions, improving onboarding time and reducing schema-related defects during integration testing.

Accenture

Hyderabad, India

Business Analyst

07/2021 - 07/2023

- Designed and built scalable ETL pipelines using Python (Pandas, PySpark) and SQL to ingest, clean, and preprocess 3 million+ daily retail transactions, supporting downstream ML and analytics workflows.
- Designed and developed scalable ETL pipelines in Python and SQL, ingesting and transforming over 5 million daily retail transactions to support downstream machine learning models and analytics initiatives.
- Collaborated with data science teams to deliver product recommendation models, increasing online conversion rates by 15% and improving customer engagement.
- Automated data quality checks, transformation workflows, and validation pipelines, reducing manual intervention by 40% and enhancing data trustworthiness.
- Optimized ETL workflows on Azure Data Factory and Azure SQL, cutting pipeline runtime by 30% and ensuring timely availability of datasets for reporting.
- Produced aggregated datasets for BI teams, enabling Power BI and Tableau dashboards to track product adoption, campaign performance, and customer retention, thereby improving decision-making speed by 25%.
- Contributed to cross-functional projects aligning machine learning recommendations with business goals, enhancing campaign ROI by 12%.
- Researched and applied data lineage, governance, and documentation standards to ensure compliance with internal and regulatory frameworks.

Experience

Phenom People	Hyderabad, India
Business Analyst Intern	04/2021 - 06/2021
- Analyzed customer adoption data across multiple product modules, identifying usage patterns that helped improve client retention by 12%. - Built SQL queries and Python scripts to extract, clean, and validate adoption metrics from relational DB, ensuring 99% data accuracy for reporting. - Developed automated Excel-based reports and visualizations to track daily client engagement trends, reducing manual reporting effort by 30%. - Collaborated with Reliability Engineering and automation teams to streamline data collection and monitoring, cutting ticket resolution time by 20%. - Prepared insights for client success teams, highlighting feature adoption gaps and driving targeted engagement campaigns that improved platform stickiness.	

Projects

Violent Crime Analysis

Analyzed crime incident and environmental data to forecast violent crime trends and support community safety planning.

- Built predictive models in R (ARIMA) achieving 85% forecasting accuracy on 50K+ records.
- Applied statistical analysis across police division and weather data to uncover correlations between external conditions and crime spikes.
- Delivered visual insights through R Shiny dashboards, enabling stakeholders to evaluate future crime patterns for resource allocation.

Cotton Plant Disease Detection using Deep Learning

Developed a deep learning solution to classify and detect cotton plant diseases for agricultural productivity.

- Implemented CNN models (VGG16, ResNet50) in Keras, achieving 87% classification accuracy. Packaged the model into a reusable pipeline.
- Applied image preprocessing techniques to enhance model reliability and reduce noise, accelerating training cycles by 15%.

Sales Analytics Dashboard with Automated ETL

Developed a centralized analytics solution to track sales performance and customer behavior for business stakeholders.

- Built ETL pipelines using SQL and Python to ingest and transform sales data from multiple sources (CRM, e-commerce, and ERP).
- Designed an interactive Power BI dashboard with dynamic filters and KPIs, improving executive decision-making speed by 25%.
- Automated daily data refresh workflows using Azure Data Factory, ensuring up-to-date metrics without manual intervention.

Certifications

Microsoft Azure Fundamentals	IBM - Data Structures & Algorithms	Python for Data Science
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